

Multiple Choice (10 marks)

1. In the diamond structure of elemental carbon, the nearest neighbors of each C atom lie at the corners of a
 - (a) square
 - (b) hexagon
 - (c) cube
 - (d) tetrahedron
2. Which of the following statements about bosons and/or fermions is true?
 - (a) Bosons have symmetric wave functions and obey the Pauli exclusion principle.
 - (b) Bosons have antisymmetric wave functions and do not obey the Pauli exclusion principle.
 - (c) Fermions have symmetric wave functions and obey the Pauli exclusion principle.
 - (d) Fermions have antisymmetric wave functions and obey the Pauli exclusion principle.
3. Which of the following is true about any system that undergoes a reversible thermodynamic process?
 - (a) There are no changes in the internal energy of the system.
 - (b) The temperature of the system remains constant during the process.
 - (c) The entropy of the system and its environment remains unchanged.
 - (d) The entropy of the system and its environment must increase.
4. By definition, the electric displacement current through a surface S is proportional to the
 - (a) rate of change of the electric flux through S
 - (b) electric flux through S
 - (c) time integral of the magnetic flux through S
 - (d) rate of change of the magnetic flux through S
5. By definition, the electric displacement current through a surface S is proportional to the
 - (a) magnetic flux through S
 - (b) rate of change of the magnetic flux through S
 - (c) time integral of the magnetic flux through S
 - (d) rate of change of the electric flux through S

True / False (10 marks)

Answer True or False

6. True or False: In the observer's reference frame, it takes 2.5 ns for the meter stick moving at $0.8c$ to pass the observer.
7. True or False: The angular magnification of the refracting telescope is 6, which is a common value for similar configurations.
8. True or False: For an electron in the $l = 2$ state, there are 5 allowed values of the magnetic quantum number m_l .
9. True or False: The sign of the charge carriers in a doped semiconductor can be determined by measuring the Hall coefficient.
10. True or False: According to BCS theory, Cooper pairs are attracted to each other by the strong nuclear force.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Discuss the relationship between the kinetic energy of electrons and their ability to diffract off a crystal surface. What is the significance of a lattice spacing of 0.4 nm in this context?
12. Discuss the final voltage across the first capacitor after it is connected to two identical, initially uncharged capacitors in series, following its disconnection from the battery.

End of Paper